Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE

Course Code & Title: CS3452 – Theory of Computation

Name of the Faculty Member(s): Mrs. S. Manjula

Innovative Practice Description

Unit / Topic: Unit V / Complexity Classes – Tractable and Intractable Problems, P and NP

Course Outcome: CO 5

Topic Learning Outcome: TLO 18

Activity Chosen: Classify the Problem Rally

Date of Activity: 07.04.2026

Venue: CSE Classroom

Justification:

- The "Classify the Problem Rally" was designed to assess and strengthen students' ability to categorise computational problems into tractable (P) and intractable (NP, NP-Complete, NP-Hard) complexity classes through active group-based reasoning and justification.
- Understanding the distinction between P and NP problems is a foundational concept in Theory of Computation that requires higher-order thinking — students must analyse, reason, and defend their classification decisions rather than simply recall definitions.
- Group-based problem classification encouraged collaborative reasoning, peer debate, and written justification of answers, deepening conceptual understanding far more effectively than passive learning.
- The rally format — where each group received specific problems and presented their classification with justification — promoted accountability, communication skills, and the ability to reason about computational complexity in an engaging and structured manner.
- By working with real-world examples of tractable and intractable problems such as Sorting, Travelling Salesman, Graph Colouring, SAT, and Knapsack, students were able to connect theoretical complexity classes to practical engineering challenges.

Time Allotted for the Activity: 40 Minutes

Details of the Implementation:

The topic focused on complexity classes — Tractable and Intractable problems, P and NP — a core concept in Unit V of Theory of Computation. The session proceeded as follows:

Step 1: Group Formation and Problem Distribution

- Students were divided into groups of 4 to 5 members each.
- Each group was assigned a set of computational problems drawn from both tractable (P class) and intractable (NP, NP-Complete, NP-Hard) categories — covering problems such as Sorting, Shortest Path, Travelling Salesman Problem (TSP), 0/1 Knapsack, Graph Colouring, Boolean Satisfiability (SAT), Hamiltonian Cycle, and Clique.
- Groups were instructed to analyse each problem and classify it into the appropriate complexity class with written justification.

Step 2: Group Discussion and Classification

- Each group collaboratively discussed the computational nature of their assigned problems and debated the reasoning behind each classification.
- Students applied their knowledge of polynomial-time algorithms, non-deterministic computation, and problem reduction techniques to support their classification arguments.
- Groups prepared clear written justifications explaining why each problem belonged to a particular complexity class.

Step 3: The Rally – Group Presentation and Peer Evaluation

- Each group presented their problem classifications and justifications to the class in a rally format — one group at a time.
- Classmates from other groups were encouraged to question, challenge, or affirm the presenting group's reasoning, creating a stimulating peer evaluation environment.
- This step simulated an academic debate setting where students had to defend their computational reasoning publicly and respond to counter-arguments.

Step 4: Faculty Verification and Consolidation

- The instructor reviewed each group's classification, acknowledged correct reasoning, corrected any misclassifications, and elaborated on the precise boundaries between P, NP, NP-Complete, and NP-Hard classes.
- The faculty consolidated the session by highlighting the significance of complexity theory in algorithm design, cryptography, and software engineering, and tied the concepts back to real-world implications.
- Students who initially struggled with boundary cases were guided through the correct reasoning, ensuring all groups concluded the session with accurate and reinforced understanding.

CO – PO / PSO Mapping:

CO	PO1	PO2	PO4	PO9	PO10	PO12	PSO1	PSO2
CO 5	2	2	1	2	1	1	2	2

1–Low 2–Moderate 3–High

PO / PSO Mapped:

Innovative Practice	PO1	PO2	PO4	PO9	PO10	PO12	PSO1	PSO2
Classify the Problem Rally	2	2	1	2	1	1	2	2

PO / PSO Justification:

PO / PSO	Justification
PO1	Students applied their knowledge of computational theory and mathematical principles to analyse the complexity of assigned problems, classifying them into tractable and intractable categories and

	demonstrating their ability to use engineering fundamentals to reason about computational solutions.
PO2	Students identified the type and structure of each assigned computational problem, formulated a classification argument using first principles of complexity theory, and reached substantiated conclusions by reasoning about polynomial-time solvability, NP-class membership, and problem reduction.
PO4	Students investigated the scope and computational nature of their assigned problems by evaluating the feasibility of efficient algorithmic solutions, analysing problem boundaries, and justifying their classification into P, NP, NP-Complete, or NP-Hard classes through systematic investigation.
PO9	Students developed problem-solving skills by collaboratively analysing a wide range of computational problems — spanning graph theory, optimisation, and satisfiability — and strengthened conflict resolution skills by debating and defending their classification justifications during the rally presentation.
PO10	Students communicated their problem classification findings effectively by presenting structured justifications to the class, interpreting technical problem descriptions accurately, and incorporating feedback from peers and the instructor to refine their understanding of complexity classes.
PO12	Students recognised the historical and ongoing significance of the P vs NP problem — one of the greatest open problems in computer science — and identified its relevance in contemporary trends such as cryptography, algorithm optimisation, and artificial intelligence research.
PSO1	Students will excel in software development, including mobile technologies, to solve complex computational tasks with soft skills by classifying computational problems as decidable/undecidable and tractable/intractable.
PSO2	Students will apply complexity theory to analyze problems, select suitable algorithmic strategies, and design efficient, scalable, and reliable software systems.

<u>Images / Screenshot of the Practice:</u>

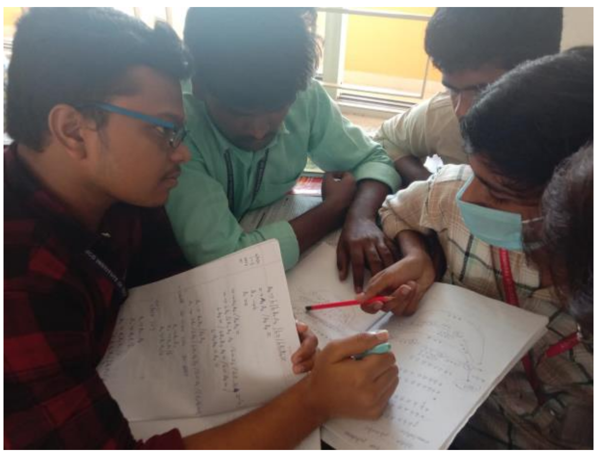

Fig 1: Students collaboratively classifying tractable and intractable problems during the rally

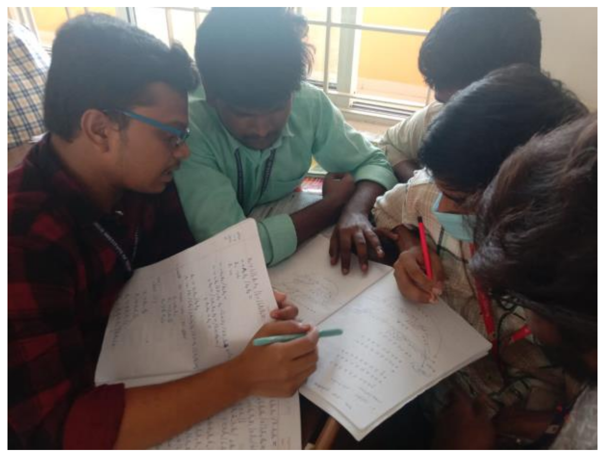

Fig 2: Group members engaged in discussion and justification of problem classifications

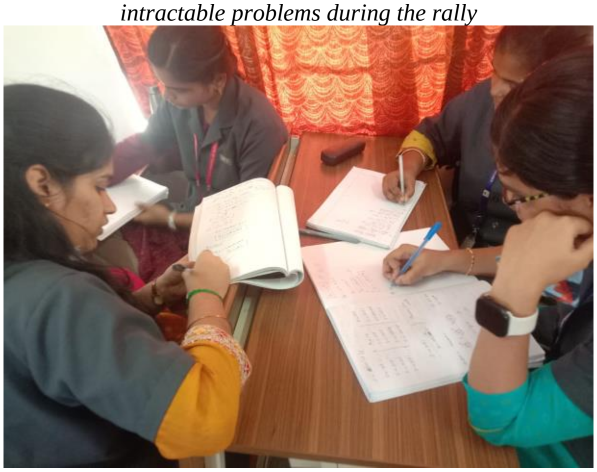

Fig 3: Students presenting their P and NP problem classifications during the rally

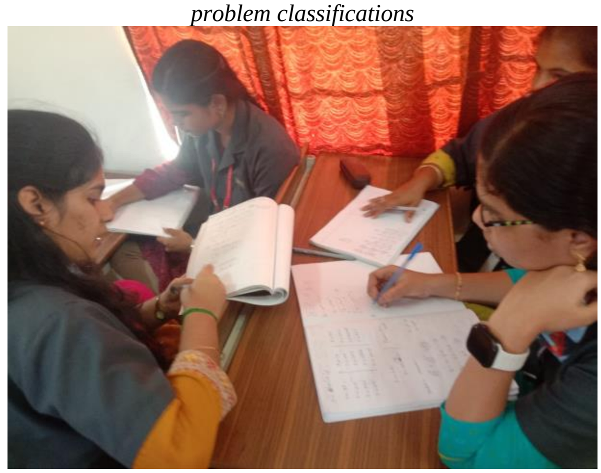

Fig 4: Active group problem analysis and written justification in progress

Reflective Critique:

Feedback of Practice from Students and Other Stakeholders:

- Students expressed that the rally format made the abstract topic of computational complexity highly engaging, helping them connect theoretical concepts with real-world problem types.
- Students appreciated the group justification exercise as it challenged them to think critically and articulate their reasoning rather than simply recalling definitions.
- Peer questioning during the rally encouraged deeper understanding and helped students identify and correct gaps in their classification reasoning.
- Faculty observed a marked improvement in students' ability to differentiate between P, NP, NP-Complete, and NP-Hard problems after the activity.

Benefit of the Practice:

- Students developed a strong conceptual understanding of complexity classes by actively analysing and justifying problem classifications rather than receiving information passively.
- The activity fostered collaborative reasoning, communication skills, and critical thinking — essential competencies for engineering problem solving.
- The rally format created a healthy and intellectually stimulating environment that motivated all groups to prepare thoroughly and defend their answers confidently.
- Outcome attainment for CO5 improved significantly as students engaged with a wide variety of tractable and intractable problems through hands-on classification and justification.

Challenges Faced in Implementation:

- Some groups initially found it difficult to distinguish between NP-Complete and NP-Hard problems, as the boundary between these classes required careful reasoning about polynomial-time reductions.
- These groups were supported by the instructor through targeted questioning and worked examples, which helped them appreciate the subtle but important distinctions between complexity sub-classes.

References:

- https://www.geeksforgeeks.org/types-of-complexity-classes-p-np-conp-np-hard-and-np-complete/
- https://www.tutorialspoint.com/design_and_analysis_of_algorithms/design_and_analysis_of_algorithms_p_np_class.htm
- https://tilt.colostate.edu/TipsAndGuides/Tip/180
- https://www.edutopia.org/topic/collaborative-learning
- https://www.ritrjpm.ac.in/images/computer-science/28.CS8501_ColloborativeLearning.pdf

Signature of Faculty Member **HOD / CSE**